

ANMOL MAHAJAN

Backend Software Engineer

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PROFESSIONAL SUMMARY

Currently pursuing a **Bachelor of Technology in Computer Science and Engineering** with strong skills in **Java** and **SQL**. Gained hands-on experience through projects focusing on distributed systems and microservices, including a solution that reduced API response latency by **76%**. Ready to contribute as an Application Engineering Intern, leveraging technical expertise in software development and systems design.

WORK EXPERIENCES

Software Development Engineer Intern | Internship

Jun 2025 - Aug 2025

Allen Digital

On-site • Bangalore, Karnataka, India

- Part of the Content and Delivery team.
- Streamlined the meeting scheduling API** end-to-end, cutting response latency from **850ms to 200ms (76% reduction)** under **500+** concurrent requests by profiling hot paths and adding targeted Redis caching.
- Redesigned the pre-session data pipeline** for over **10,000 daily active users**, reducing page load time by **35%** through selective fetching and Redis-backed caching.
- Eliminated **3 redundant message queue call patterns** across join, leave, and resource flows, cutting queue overhead by **40%** and lowering infrastructure cost.
- Introduced stream-wise filtering and flag-based routing across **8+** API paths, reducing irrelevant push notification delivery by **60%**.
- Refactored the Join Meeting workflow in a microservice handling over **1 million monthly events**, removing **200+** lines of dead code and improving maintainability.

Technologies / Skills Used : Go, Redis, Messaging queues, SQL

PROJECTS

DistriLimit – Multi-Node Rate Limiting Gateway | GitHub

2026

- Eliminated** distributed rate-limit races by centralizing usage state in **Redis** and using **Lua scripts** for atomic updates across server instances.
- Validated** global rate limits across **3 concurrent gateway instances** sharing Redis, ensuring requests never exceeded the configured limit due to instance scaling.
- Implemented and load-tested 4 rate-limit algorithms** with **k6**, benchmarking Token Bucket, Leaky Bucket, Sliding Window Log, and Counter performance.
- Built** an async **Kafka** → **PostgreSQL** audit pipeline and database-driven client configuration, enabling request logging and limit changes without redeployment.

Technologies / Tools Used : Spring Boot, Redis, Kafka, PostgreSQL, SQL

Gopherflow – DAG-Based Workflow Orchestration Engine | GitHub

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Technologies / Tools Used : Spring Boot, Kafka, Redis, PostgreSQL, Python

TT Platform | GitHub Apk

2025

- Shipped** a full-stack tournament platform with **React Native + Spring Boot**, supporting **50+ users, 20+ tournaments, and 500+ matches** on Android.
- Delivered** real-time scoring via **WebSockets**, processing **100+ live events/hour** with automated round-robin scheduling, Elo rankings, and MVP awards.
- Secured 25+ REST APIs** with **OAuth2, email OTP, JWT, and RBAC**, while optimizing Render deployment to achieve **<3s p95 latency**.

Technologies / Tools Used : Spring Boot, PostgreSQL, WebSockets, Firebase, JWT

SKILLS

Databases :	MySQL, Redis, PostgreSQL
Soft Skills :	Agile/Scrum, Object-Oriented Programming, Data Structures and Algorithms
Tools & Platforms :	Docker, Git, Prometheus, Grafana
Frameworks & Libraries :	Spring Boot, Redis, Kafka
Programming Languages :	Java, Go, SQL, C++

EDUCATIONS

Guru Nanak Dev University | Amritsar, Punjab

Jan 2023

Bachelor of Technology in Computer Science and Engineering

CGPA : 8.03/10

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